■ Eco_Road: Clean-Air Student Navigation

12-Hour Hackathon Blueprint & Implementation Roadmap | 3-Developer Team Strategy

Sprint Time: 12 Hours Total	**Team Size:** 3 Members	**Target Audience:** Campus Students & Walkers
Base Repo: GANESH-NADKARNI/AQI-Navigation	**Core Problem:** Toxic Inhalation on Campus	**Key Goal:** Live Turn-by-Turn Google Maps UX

1. Problem Statement & Hackathon Mission

The Crisis: Millions of university students and urban pedestrians walk through toxic smog corridors, busy bus terminals, construction choke-points, and high-emission arteries daily. Most navigation apps (like standard Google Maps) optimize strictly for *duration* or *shortest vehicular distance*, inadvertently funneling students through the heaviest particulate matter (PM2.5/PM10) zones.

The Solution (Eco_Road): An interactive, real-time walking navigation system designed for campus students. It calculates dual routes (Fastest vs. Cleanest Air Route) using real-time open-source Air Quality Index (AQI) data. Crucially, it provides a full **Google Maps mobile navigation experience** (turn-by-turn HUD, heading rotation, real-time re-centering, and voice guidance) alongside quantifiable **Respiratory Exposure Savings** (e.g. 'Saved 42% PM2.5 intake').

2. Base Repository Audit & Key Deficiencies to Solve

Your team is starting with the GANESH-NADKARNI/AQI-Navigation codebase. Here is an honest audit of what it does well and what must be upgraded to win the hackathon:

Subsystem	Current Repo State	The Critical Problem	12-Hour Winning Upgrade
Routing Profile	Calls OSRM /driving/ API	Routes via expressways/highways; ignores campus pedestrian walkways & footpaths.	Switch to OSRM /foot/ or walking geometry mode for authentic student paths.
Navigation Experience	Basic 2D top-down GPS dot with Leaflet circle	No bearing compass tilt, no smooth animation; GPS jumps erratically.	Implement smooth marker interpolation, bearing cone, auto-tilt & sleek turn HUD.
Judge Demoability	Only live GPS watchPosition	Judges inside a hall cannot walk outside! Live GPS will show 0 speed or fail.	**Walk Simulator Mode:** Playback slider (1x, 2x, 5x) simulating real-time walk along the route.
Health Metrics	Single average AQI number (e.g. '142 Moderate')	Abstract number; students don't know biological impact or why to care.	Calculate **Inhaled Micrograms of PM2.5** and **Cigarette Equivalence avoided**.
Visual Feedback	Single solid blue/green line with a few dots	Cannot see which specific streets or intersections are polluted.	Segmented polyline with color gradient (Green → Yellow → Red along street blocks).
Data Reliability	Direct calls to WAQI API only	WAQI rate-limits or lacks dense sensors inside university campuses.	Synthetic/IDW (Inverse Distance Weighting) Campus interpolation & mock fallback cache.

3. Team Division: 3 Developers, 1 Mission

To execute this within 12 hours without blocking each other, the 3 developers must work on distinct, loosely coupled pillars:

Team Member	Core Domain	Key Responsibilities & Deliverables	Tools & Tech
Member 1 *(Lead Frontend & Map UX)*	**Google Maps UI & Dynamic Navigation Engine**	• Turn Leaflet map into true Google Maps experience. • Heading compass, smooth user marker interpolation (no jerky jumps). • Dynamic auto-re-centering with user manual-drag unlock. • **Walk Simulation Controller:** Demo slider for judges to simulate walking at 1x/2x/5x speed. • Turn HUD card (Next turn icon, distance countdown, road name). • Voice navigation triggers (Web Speech API) as simulated/live user approaches turns.	Leaflet.js HTML5 Geolocation CSS3 Transforms Web Speech API
Member 2 *(Backend & Routing Engine)*	**Pedestrian Routing & AQI Exposure Analytics**	• Switch OSRM routing from driving to pedestrian/walking foot mode. • Build Route Cost Function: Cost = Distance + λ × AQI_penalty. • Real-time AQI Dosage Calculator: Inhaled PM2.5 in μg = Σ(AQI × Time × Inhalation Rate). • Multi-point polyline segment colorizer (split route into green/yellow/red blocks). • Bulletproof API caching & robust mock fallbacks so demo NEVER fails if WAQI rate-limits.	FastAPI (Python) OSRM API WAQI / OpenAQ NumPy / SciPy
Member 3 *(Product, Health & Pitch Lead)*	**Student Health Features, Campus Hotspots & Pitch Deck**	• Student Health Persona mode: 'Standard Walker' vs. 'Asthma / Respiratory Sensitive'. • Campus Hotspot Overlays (e.g. Bus Terminal, Cafeteria Exhaust, Green Quad). • 'Respiratory Impact Card': Compare Fastest vs Clean Route in PM2.5 savings and avoided cigarettes. • Real-time In-Route Smog Alert popup ('Entering heavy traffic zone, recommend detour'). • Create 6-slide winning pitch deck, live demo script, and 90-second backup demo video.	Tailwind / Modern CSS Canva / Slides Screen Recorder Product Design

4. 12-Hour Sprint Breakdown (Hour-by-Hour Roadmap)

A strict, milestone-driven hourly roadmap to guarantee delivery with zero last-minute panic:

Phase & Hours	Member 1 (Frontend & Navigation)	Member 2 (Backend & Routing)	Member 3 (Health & Presentation)	Milestone Deliverable
Hours 00–02 Setup & Base Audit	Clone repo, run frontend locally. Extract navigation UI into full-screen mobile app layout.	Set up FastAPI backend. Verify OSRM pedestrian endpoints. Implement mock AQI data fallback.	Map campus/city hotspot coordinates. Draft user personas (student walking to library/hostel).	**Working local build with pedestrian route lines rendering.**
Hours 02–04 Core Navigation & Algorithms	Build **Walk Simulator Mode** (interpolated step runner, speed slider 1x-5x) & smooth marker.	Implement Clean-Air Route Scoring Algorithm. Calculate AQI segment weights along coordinate polyline.	Design 'Health Impact / Lung Exposure' UI cards and asthma vulnerability toggle.	**Simulated dot moves along route; backend scores dual routes with clean metrics.**

Hours 04–06 Google Maps Polish & Exposure	Add Google-Maps style rotating heading cone, dynamic camera follow, and turn countdown banner.	Build Respiratory Dosage formula: calculate PM2.5 µg inhaled based on duration and AQI.	Add Campus Hotspots (Bus bay, cafeteria smoke zone) geofences and visual alert banners.	**Interactive Google Maps look & feel with live distance and dosage counter.**
Hours 06–08 Deep Integration & Heatmaps	Render multi-colored polyline segments (green for clean street, red for smog choke-point).	Integrate dynamic rerouting trigger: notify frontend if high AQI zone is approached.	Build 'Eco vs Fast Summary Comparison' modal (Time diff vs Pollution saved).	**Full end-to-end user journey: Search → Select Eco → Live Turn Nav.**
Hours 08–10 Hardening & Edge Cases	Audio voice-turn cues polish. Re-center button behavior. Mobile responsive viewport test.	Cache all route responses locally to ensure instant response times during demo.	Create pitch slide deck (Problem, Architecture, Live Demo, Health Impact, Future Scale).	**Robust application that works 100% offline or online without crashes.**
Hours 10–12 Demo Rehearsal & Pitch	Final UI styling fixes, sound toggles, high-contrast dark/clean-air mode.	Assist demo flow. Prepare backend terminal metrics display if judges ask tech questions.	Record 90-sec backup video demo. Rehearse pitch 3 times with live walk simulation.	**Submission Ready: Code committed, video recorded, slide deck finalized.**

5. Technical Upgrades & 'Secret Sauce' Features

To stand out to hackathon judges, Eco_Road implements three killer differentiators that standard student projects miss:

■ ***Secret Weapon 1: The 'Judge Walk Simulator' (Zero Demo Risk)***
The Problem: In hackathons, teams demoing GPS navigation fail because they are indoors under concrete roofs where GPS signal drifts wildly or reports 0 km/h.
The Implementation: In navigation.js, implement a dual-mode engine: ***Live GPS*** *vs.* ***Simulation Mode****. In Simulation Mode, the system steps through route waypoints with a customizable slider (1x walking speed 1.2 m/s, up to 5x fast-forward). As the simulated position moves, the camera pans smoothly, heading rotates, remaining distance drops, and voice turns trigger at 30m. Judges can watch the full Google Maps navigation experience unfold live on screen without stepping outside!*

■ ***Secret Weapon 2: Scientific Respiratory Exposure Dosage***
The Problem: Showing 'Route A: AQI 180' vs 'Route B: AQI 120' feels abstract. Students don't grasp the real health danger.
The Scientific Formula: Pedestrian Minute Volume is ~12 Liters/min (walking pace = 0.012 m³/min).
Total PM2.5 Inhaled (μg) = ∑ [Concentration_i (μg/m³) × Walking_Time_i (min) × Breathing_Rate (0.012 m³/min)]
The User Impact: Instead of just 'Clean Route', Eco_Road displays:
'Eco-Route saves 38.4 μg PM2.5 inhalation (~0.4 cigarettes avoided) for just 3.5 extra walking minutes.' *This directly satisfies the hackathon challenge requirement regarding respiratory health impact.*

■ ***Secret Weapon 3: Campus 'Clean Air Corridors' & Sensitive Mode***
The Implementation: Include a dedicated toggle for ***'Campus Mode'*** *with pre-mapped university hot-spots (Main Gate diesel bus exhaust, Cafeteria waste incineration, Central Botanical Park green buffer). Users can toggle* ***'Asthma / Sensitive Respiratory Profile'****, which increases the AQI penalty multiplier (λ) in route selection, guaranteeing routing through green tree-lined campus corridors rather than perimeter roads with heavy vehicle traffic.*

6. Architecture & Implementation Specification

The system maintains a clean decoupling between the client UI and the calculation backend:

Module / File	Current State	Target Implementation (Sprint Changes)
frontend/js/navigation.js	Simple watchPosition callback	Add Simulation engine, lerp interpolation for marker heading & coordinates, turn distance countdown.
frontend/js/map.js	Leaflet static tile view	Add bearing rotation/tilt wrapper, colored segmented polylines, pulsing blue location halo.
backend/services/routing_service.py	Hardcoded driving OSRM endpoint	Add walking profile support, cost function scoring, and microgram inhalation dosage calculations.
backend/services/aqi_service.py	Direct WAQI/OWM queries	Add offline cache, campus hotspot bounding boxes, and fallback synthetic AQI interpolator.
frontend/index.html & css	Standard dark web page layout	Refactor into Google Maps mobile app shell: top search pill, bottom swipe card, floating recenter button.

7. Hackathon Pitch Deck & Live Demo Script

A brilliant product loses if the pitch is disorganized. Follow this 3-minute winning presentation structure:

Slide / Stage	Time	Core Message & Visual	Presenter Action
Slide 1: Hook	0:00 - 0:30	*'Students breathe the equivalent of 3 cigarettes walking to class.'* Show photo of student next to bus exhaust.	Member 3 states problem vividly with emotion and campus relevance.
Slide 2: The Flaw in Maps	0:30 - 0:50	Google Maps prioritizes the fastest highway. It is completely blind to air toxicity.	Explain why current navigation tools fail pedestrians.
Live Demo: Eco_Road	0:50 - 2:00	1. Type Start/End destination. 2. Dual Routes appear: Shortest (12 min, AQI 240) vs Eco (15 min, AQI 65). 3. Tap 'Start Navigation' → Walk Simulator starts moving. 4. Map follows marker, turn arrow updates, voice warns of upcoming clean turn.	Member 1 & 2 run live interactive simulation on screen.
Slide 3: Impact & Tech	2:00 - 2:35	Highlight the Respiratory Dosage formula, open-source stack (OSRM + Leaflet + WAQI), zero API cost.	Member 2 explains algorithmic routing cost penalty.
Slide 4: Future Vision	2:35 - 3:00	Campus IoT air-sensor integration, wearable smart watch vibration alerts, municipal green zones.	Member 3 closes with memorable call to action for clean student health.

8. Hackathon Contingency Matrix (Preventing Disasters)

Hackathons have unpredictable network outages and API throttles. Here is how your team stays bulletproof:

Risk Event	Severity	Immediate Contingency Plan
WAQI / OWM API Rate Limit / Network Down	**CRITICAL**	Backend automatically serves pre-cached synthetic AQI grids based on coordinates if external API fails.
OSRM Public Routing API slow or down	**HIGH**	Pre-bundle 3 popular campus route GeoJSONs locally in frontend as fallback instant demo routes.
Browser Blocks Web Speech Audio	**MEDIUM**	Web Speech requires user gesture: trigger an audio test chime on the very first 'Get Routes' button click.
Judges ask to test a custom city	**LOW**	Nominatim geocoding handles any global city; fallback interpolator guarantees clean AQI scoring anywhere.

Summary Checklist for Success: Focus on visual polish and demo stability. A working walk-simulator and a clear respiratory health savings metric will score higher than complex theoretical models. Stick to your assigned roles, meet at the 2-hour milestone checkpoints, and maintain code freeze at Hour 10.